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Wide area coverage Raman spectroscopy for reliable quantitative analysis and its applications

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This review summarizes recent studies to improve sample representation in Raman measurement by covering a large area of a sample in spectral collection. Three different schemes have been mainly investigated to fulfill the goal: (1) averaging of Raman spectra collected at many different locations on a sample, (2) rotation of a sample during spectral collection and (3) simultaneous wide area illumination (WAI) for spectral collection. The use of a wide area illumination scheme, simultaneously illuminating a laser over a large area for spectral acquisition without any further assistance such as sample rotation, has increased in diverse fields. Applications employing the WAI scheme in pharmaceutical, polymer/chemical/petrochemical and other areas are described in this review.

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Introduction

Since the mid-1980s, the demand for fast, non-destructive analytical methods for quality control (QC) and quality assurance (QA) has substantially increased in order to maximize the efficiency of production in many industrial fields. Near-infrared (NIR) spectroscopy[1–3] has received great interest from many analysts since it can readily fulfill the requirement of rapid quality monitoring. The popularity of NIR spectroscopy was mainly driven by its combination with chemometrics based on diverse multivariate calibration methods that established correlations between NIR spectra and the corresponding concentration/property variations.[4–7] Without combining with multivariate analysis, NIR spectroscopy would be of little value

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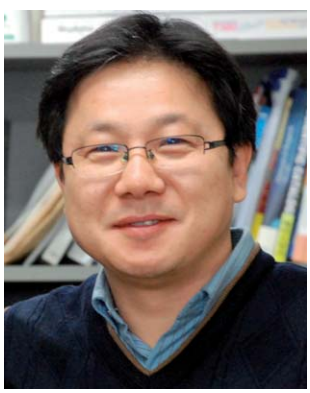

Hoeil Chung is a Professor in the Department of Chemistry in Hanyang University, Seoul, Korea. He obtained his B.S. from Hanyang University in 1989 and Ph.D. from the University of Iowa in 1994. Right after obtaining his Ph.D., he joined SK Energy (formerly Yukong) in Korea as a research scientist and was involved in projects for developing diverse near-infrared (NIR) spectroscopic systems for on-line monitoring of refinery and petrochemical processes; for example gasoline blending and naphtha reforming. In 2002, he joined Hanyang University and started his academic research in the areas of transmission/wide area coverage Raman spectroscopy for non-destructive analysis, reproducible and reusable SERS schemes to detect disease biomarkers, electrochemical synthesis of metal–alloy nanostructures and development of advanced chemometric algorithms. He has published more than 100 papers in peer-reviewed journals and has presented them at many international conferences.

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in analytical chemistry since NIR spectral features are usually indistinct and highly overlapping due to the resultant overtone and combination vibrations, thus providing limited information on the molecular structure and composition.

The vogue of NIR spectroscopy in this period also triggered the evaluation of Raman spectroscopy for the same purpose, since it is also capable of simple, non-destructive analysis with better-resolved spectral features. Raman spectral features are substantially more distinct compared to NIR bands and vary sensitively according to the molecular structure and symmetry. However, it has been relatively less exploited due to several drawbacks such as sub-sampling problems, occasional fluorescence hampering the accuracy of analysis and expensive instrumentation. Among them, the most intrinsic issue in Raman measurement is the possible sub-sampling that eventually results in a non-representative spectral acquisition of a sample. For Raman spectral collection in general, the laser irradiates only a small volume of the sample such as a few hundred micrometers in diameter, so the acquired spectrum is only representative of a localized area of the sample rather than the whole.[8,9] For reliable quantitative analysis using Raman spectroscopy, the occurrence of sub-sampling should be eliminated or minimized at least. The most adopted strategy to overcome the problem is to increase the sample coverage volume for Raman spectral collection, thereby achieving an improved or correct compositional representation of a sample.

In this publication, we have reviewed studies that made an effort to improve sample representation using wide area coverage schemes. Three different schemes, averaging of Raman spectra collected at different locations, rotation of samples during spectral collection and wide area illumination (WAI), have been mostly employed. Each scheme is described and their applications are categorized into three fields: pharmaceutical, polymer/chemical/petrochemical and others.

Finally, a brief perspective of wide area coverage Raman measurement is provided in the Conclusions section.

Schemes to overcome the sub-sampling problem in Raman measurement and their applications

To increase the reliability of Raman spectroscopy for quantitative analysis of various kinds of samples, inaccurate compositional representation of samples originating from sub-sampling needs to be resolved. It is especially problematic when inhomogeneities as well as spatial variations in composition are present in the sample matrix.[10–12] The sub-sampling problem has been investigated in detail by several researchers. Taylor *et al.*[8,10] demonstrated the need for homogeneous calibration/validation samples for robust quantitative analysis and discussed the difficulties associated with collecting a Raman spectrum truly representative of the concentration of samples. Campbell Roberts *et al.*[8,11] investigated day-to-day reproducibility and intra-day reproducibility in quantitative analysis in relation to sample mixing, sampling positioning, laser power and particle size. The largest sources of error were sample inhomogeneity and small sampling area. In addition, they demonstrated that the use of a sample with smaller particle sizes improved the homogeneity of the sample, yielding a better Raman spectroscopic representation of the sample as well as a more accurate analysis. Bell *et al.*[8,12] recorded Raman spectra from a grid of points on individual pharmaceutical tablets by stepping a stage in the *x* and *y* directions. Then, the spectra were individually examined as well as compared with an average spectrum. Three major components of the tablet, MDMA (*N*-methyl-3,4-methylenedioxyamphetamine), caffeine and lactose, were inhomogeneously distributed on a microscopic scale, so representative sampling was critical.

As demonstrated in the publications, the coverage of a larger sample volume in Raman spectral collection is a straightforward approach to ensure the correct spectral representation of a sample. For this purpose, three major schemes have been developed, as shown in Fig. 1: averaging of Raman spectra collected at many different locations on a sample (a), rotation of a sample during spectral collection (b) and simultaneous wide area illumination for spectral collection (c). All of the schemes were used to increase the sample coverage volume for Raman

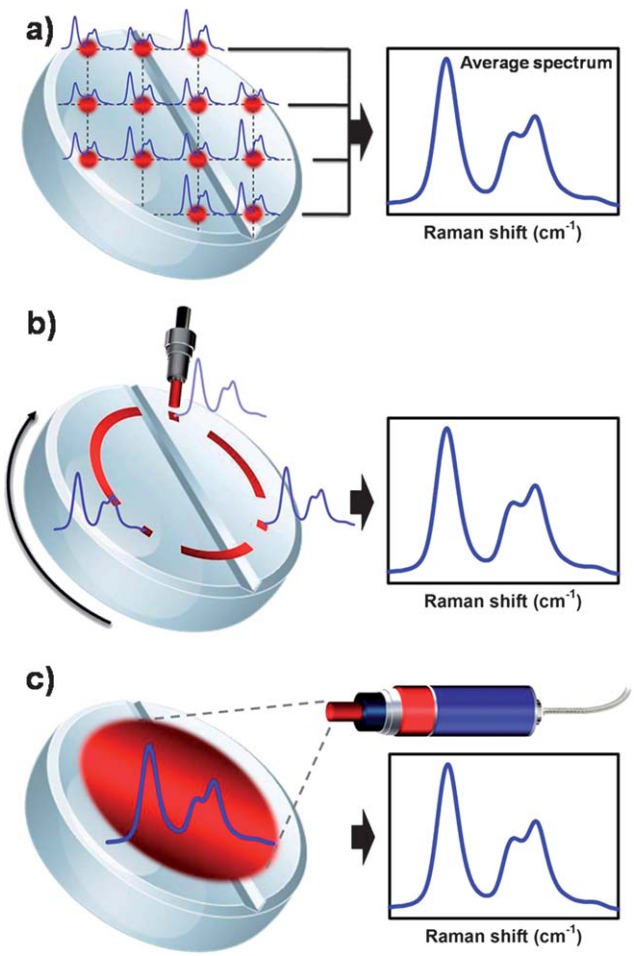

Fig. 1 Three major schemes to improve sample representation through wide area coverage of a sample: (a) averaging of Raman spectra collected at many different spots on a sample, (b) rotation of a sample during spectral collection and (c) simultaneous wide area illumination for spectral collection.

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spectral collection. Details of each scheme are described in the following sections.

(1) Averaging of Raman spectra collected at many different locations on a sample

The simplest and most basic way to overcome or lessen sub-sampling problems is to average the sub-spectra acquired from numerous sample locations. Then, the averaged spectrum could be used to achieve a representative quantitative analysis. In practice, manual sequential acquisition of Raman spectra at different locations is unrealistic. Therefore, several groups have developed devices enabling automatic spectral acquisition (or mapping) over a large area. Bell *et al.*[12] reported a scheme capable of collecting Raman spectra at multiple points using a motorized sample stage in the x and y directions. The use of spectra averaged from the largest number of points provided the lowest sampling error for the measurement of sample composition. They claimed that the optimum solution was to acquire spectra at the smallest number of points ensuring an appropriately low sampling error. An obvious way to reduce sampling error in Raman measurement is to increase the number of sampling points; however, this substantially lengthens the analysis time and practically limits its application in routine analysis, as large numbers of samples need to be analyzed in a short period of time.

To acquire Raman spectra at different locations in a much faster fashion, the methods of line illumination[13,14] and global Raman illumination[14–20] have been demonstrated. In line illumination, a laser line is raster scanned along either the x or y axis with full height slit binning on a CCD. It enables the measurement of multiple points at once and individual spectra collected along the line are combined together to produce an average line spectrum when the CCD is binned. In global Raman illumination, a widened laser beam illuminates a sample (typically 20–400 μm in diameter) and the scattered radiation generated from the covered area is delivered to the CCD through a selection device such as a liquid crystal tunable filter (LCTF). Data collection is typically at pre-selected wavenumbers, so full-range spectra are not normally collected. Thus, the CCD observes only a discrete wavenumber slice of the overall spectral range, like a camera recording a snap-shot through a color filter. Therefore, global Raman illumination has been mostly utilized to generate Raman images for pharmaceuticals and polymers.[14–20] Further, when acquisition of a spectrum with full range is in demand, the LCTF must be tuned to select different wavelengths in a continuous manner. Then, individual spectra are averaged to improve the sample representation. However, it leads to a significant increase in collection time and so dilutes the major advantage of fast imaging capability.[17,19] Also, the spectral resolution is generally 10 cm^{-1}, so it may be insufficient when recognition of a minor spectral difference is critically required.[17,19]

(2) Rotation of a sample during spectral collection

Rather than collecting spectra at many different spots, the rotation of a sample with fixed laser illumination to increase the sample coverage area is more advantageous in terms of the speed of spectral acquisition and simplicity in instrumentation. Meanwhile, overheating of an irradiated spot occasionally occurs when the laser illuminates the same spot over a longer period of time. Using sample rotation, sample overheating or burning can be reduced.[21] Several groups have described improved quantitative Raman analysis with the incorporation of sample rotation.[21–32] Langkilde *et al.*[22] reported an experimental setup where a powdered sample was placed in a NMR tube. It was rotated and moved vertically during the spectral collection. This setup led to coverage of a much larger sampling volume, thus reducing the errors in the analysis of inhomogeneous mixtures. Interestingly, Johansson *et al.*[23] investigated the effect of different irradiance patterns on the accuracy of quantitative analysis for intact immediate-release tablets that were rotated during spectral collection. Point irradiation resulted in the largest error, followed in order by small circle, large circle and area irradiation. In addition, they reported that the use of multivariate analysis such as partial least squares (PLS) was helpful to improve the accuracy of concentration determination of tablets compared to using univariate analysis. Hwang *et al.*[24] also compared Raman spectra collected at different spots on an ambroxol tablet with those acquired with tablet rotation and ultimately demonstrated the analytical utility of sampling rotation.

When a high-throughput and/or on-line analysis requiring continuous measurement of many samples is necessary, the rotation of samples during spectral collection cannot quench the demand. Moreover, the use of a peripheral accessory for sample rotation is still quite cumbersome. Therefore, a Raman collection scheme that is able to synchronously cover a large sample area without either sample rotation or averaging of Raman spectra collected at different spots is in great demand.

(3) WAI scheme

The most efficient scheme to acquire representative Raman spectra of a sample is simultaneous wide area irradiation of laser on a sample to cover a large sample volume without any further assistance such as sample rotation. In the case of using FT-Raman instrumentation, a large circular aperture to create a laser beam with a size of up to 0.5–1 mm is readily available; therefore, Raman spectra with a better sample representation can be collected.[33–35] The review paper claimed that the good reproducibility in FT-Raman spectroscopic experiments was largely attributed to the large entrance aperture, allowing laser to be focused on a relatively large sample volume.[33]

One of the milestones in the development of the simultaneous wide area coverage scheme based on dispersive instrumentation is the PhAT probe developed by Kaiser Optical Systems (Ann Arbor, MI, USA).[36,37] This probe provides a 6 mm diameter laser spot with a focal length of ~25.0 cm as well as a 3 mm diameter laser spot with a focal length of ~12.5 cm. The excitation laser (785 nm, diode laser) was introduced using an array of 50 optical fibers and magnified to form a large circular illumination area. Then, the scattered Raman signal was collected by the same optical fibers and delivered to a CCD detector. Moreover, the

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PhAT probe took advantage of the efficient signal collection resulting from non-confocal global illumination, while binning and co-addition of signals from individual fibers were used to minimize the spectral acquisition time.

The utility of the PhAT probe, alternatively known as the WAI scheme, was largely exploited by Taylor's research group.[36] They prepared three commercial probes with different illumination spot sizes (60 μm diameter of immersion optics, 150 μm diameter of standard non-contact optics (NCO) and 3 mm diameter of the WAI scheme) and estimated the penetration depth in each case. By increasing the thickness of anhydrous theophylline pellets, the penetration depth was determined as the thickness when the peak intensity at 1707 cm^{-1} started to level off. They found that the penetration depths using 60 μm, 150 μm and 3 mm diameter spots were about 300 μm, 600 μm and 2 mm, respectively. Overall, the sampling volume of the WAI scheme was approximately 1300 times larger than that for the 150 μm diameter probe.

Using the same three probes, they also assessed the variation of Raman intensity when the distance between the sample (anhydrous theophylline powder) and the tip of the probes changed. When using the WAI scheme, the signal was maximized around the focal length of 18 cm, and the variation of Raman intensity was much less sensitive to the change of distance compared with those using the small-spot schemes.

Chung's research group extensively demonstrated the analytical usage of the WAI scheme for quantitative analysis of samples in diverse fields such as pharmaceuticals,[37–43] polymers,[44–48] petrochemicals[49] and agricultural samples.[50] In their studies, the WAI scheme was combined with a synchronous intensity correction standard to compensate for the problematic variation of Raman intensity from the change of laser power to enhance the performance of quantitative analysis.[37–39,51–54]

Due to the superior performance of WAI-based Raman measurement, it has been widely applied for analysis of samples in many different fields. In this review, we have summarized the applications into three categories: (a) pharmaceutical, (b) polymer/chemical/petrochemical and (c) other areas, such as agriculture and cosmetics. For an easy and clear presentation, the applications are summarized in relevant tables (refer to Tables 1 and 2) with a brief description of each application.

(a) Applications in pharmaceutical analysis. The extensive use of Raman spectroscopy in the pharmaceutical field was triggered by the announcement of the process analytical technology (PAT) guideline by the Food and Drug Administration (FDA) in 2004, requiring a substantial increase in the investigation of new and rapid analytical methodologies for at-line and on-line measurements.[55] PAT is an evolving philosophy in the pharmaceutical industry that promotes strategies for the control of primary and secondary manufacturing processes. Raman spectroscopy is one of the potential analytical methods to meet the requirement, so large numbers of publications that utilized the scheme of wide area coverage for pharmaceutical analysis have been recently reported. Table 1 summarizes the applications of the WAI scheme in the pharmaceutical field. A brief summary of each application is provided, and the applications are grouped by research group.

Chung's group initially demonstrated the reliability of the WAI scheme for the compositional analysis of active pharmaceutical ingredients (APIs) in naproxen tablets.[37] Simultaneously, the effective and synchronous standard configuration using isobutyric anhydride was incorporated to correct for the problematic variation of Raman intensity occasionally occurring from the change of laser power. The prediction accuracy based on PLS regression was not sensitive to changes in the laser power or tablet position within ±2 mm. Further, by utilizing the lower sensitivity of the WAI scheme with respect to sample placement due to the long focal length, they directly determined the API concentration of pharmaceutical samples in diverse containers, such as ambroxol in a capsule,[42,43] povidone in a squeezable LDPE bottle[39] and H_2O_2 in a PE bottle.[40] In addition, using Raman microscopy, they investigated the possible micro-inhomogeneity in an acetaminophen suspension[38] as well as urea cream[41] that are normally regarded as quite homogeneous samples for Raman measurement. They found that even micro-inhomogeneity present in these samples could possibly degrade the accuracy of quantitative analysis when the Raman sampling volume was small and suggested a large area coverage in the spectral collection for reliable analysis of samples with micro-inhomogeneity.

Taylor's group mainly investigated wide area coverage Raman measurement for monitoring pharmaceutical processes such as wet granulation[36,57] and crystallization.[58] Since a solvent-mediated phase conversion of anhydrous theophylline to the monohydrate form occurs rapidly during aqueous wet granulation, a fast in-line analytical method to follow the transformation quantitatively is necessary. The accuracy for determining the amount of transformed theophylline using the WAI scheme was significantly better than that acquired using the standard spot-size (60 μm) scheme used in their previous publications.[36,59] With the use of in-line Raman measurement, they also monitored the phase transformations of three model compounds (caffeine, carbamazepine and sulfaguanidine) during a high shear wet granulation, performed in the presence and absence of various polymeric excipients, to determine the extent of the inhibitory effects.[57] The key for successful monitoring of the phase transformation was mainly based on the collection of representative Raman spectra of samples in the process using wide area coverage.

Kleinebudde's group was interested in determining the amount of coating and monitoring the active coating on diprophylline tablets.[60–62] Initially, they performed a scale-down experiment and showed the transferability of a PLS model developed on a mini-scale coater to application on a micro-scale coater. Then, they developed a multivariate quantitative calibration model using the spectra acquired at different stages of coating from a small-scale pan coater. The developed model was tested to monitor the progress of coating. In addition, the predicted weight gain and the amount of coated diprophylline under the conditions of different working distances were not significantly different from each other, demonstrating the possibility of in-line monitoring of the coating process within the working distance range of 15 to 25 cm.

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Table 1 Summary of applications of the WAI scheme in Raman spectroscopic analysis for pharmaceutical samples

Type of analysis	Sample type (formulation)	API	Analytical feature	Reference
Determination of API concentration	Tablet	Naproxen	Non-destructive analysis of tablets with an external standard	Chung *et al.*[37]
Determination of API concentration	Turbid suspension	Acetaminophen	Determination of API concentration in a matrix of turbid suspension	Chung *et al.*[38]
Determination of API concentration	Liquid in a plastic (LDPE) bottle	Povidone	Direct measurement of API through a squeeze bottle	Chung *et al.*[39]
Determination of API concentration	Liquid in a plastic (HDPE) bottle	Hydrogen peroxide	Demonstrated reproducible spectral collection under minor variations of bottle positions	Chung *et al.*[40]
Determination of API concentration	Cream	Urea	Investigated micro-inhomogeneity in a cream and demonstrated the necessity of wide area coverage for reliable quantitative analysis	Chung *et al.*[41]
Determination of API concentration	Capsule	Ambroxol	Determination of API concentration in a capsule directly	Chung *et al.*[42]
Determination of API concentration	Capsule	Ambroxol	Determination of API concentration in capsules with different colors	Chung *et al.*[43]
Determination of API concentration	Powder	Nitrofurantoin, *etc.*	Investigated the effect of sampling procedure and instrumental configuration on the model prediction	Taylor *et al.*[56]
Process monitoring (wet granulation)	Powder	Theophylline	Assessed homogeneity in powder blends and investigated changes in the solid state form during wet granulation	Taylor *et al.*[36]
Process monitoring (wet granulation)	Powder	Caffeine, *etc.*	Assessed the ability of various polymeric excipients to inhibit the anhydrate to hydrate transformation during high shear wet granulation	Taylor *et al.*[57]
Process monitoring (crystallization)	Solid/liquid	Atenolol, *etc.*	Detection of amorphous phase and different polymorphs	Taylor *et al.*[58]
Process monitoring (coating)	Tablet	Diprophylline	Prediction of the coated amount of API during the coating process on a mini-scale and a micro-scale pan coater	Kleinebudde *et al.*[60]
Process monitoring (coating)	Tablet	Diprophylline	In-line quantitative monitoring of active coating	Kleinebudde *et al.*[61]
Process monitoring (coating)	Tablet	Diprophylline	In-line evaluation of the effect of process parameters and coating uniformity	Kleinebudde *et al.*[62]
Process monitoring (blending)	Powder	Aspirin, *etc.*	*In situ* monitoring of powder mixing in a high shear blender and investigation of the effect of particle size on the sampling depth	Littlejohn *et al.*[63]
Process monitoring (drying)	Powder	Cellobiose octaacetate	*In situ* measurements to produce drying curves based on direct analysis of the particle properties	Littlejohn *et al.*[65]

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Table 1 (*Contd.*)

Type of analysis	Sample type (formulation)	API	Analytical feature	Reference
Process monitoring (crystallization/drying)	Solid/suspension	SAR474832	Coupling of Raman spectroscopy with dynamic water vapor sorption gravimetry (DWVSG) for the monitoring of phase transition	Feth *et al.*[66]
Process monitoring (crystallization/drying)	Powder/suspension	SAR474832	On-line study of drying, crystallization, isolation, and micronization	Feth *et al.*[67]
Forensic investigation	Tablet	Bromazepam, *etc.*	Developed a calibration strategy for automatic identification of the tablets (25 different product families)	Roggo and Dégardin *et al.*[68]
Forensic investigation	Tablet/capsule	Sibutramine, *etc.*	Identification of 31 types of genuine capsules and tablets, and detection of their counterfeits	Roggo and Dégardin *et al.*[69]
Forensic investigation	Tablet	Ciprofloxacin	Identification of counterfeit medicines by laboratory *versus* handheld instruments	Moffat *et al.*[70]
Process monitoring (self-emulsifying drug delivery systems, SEDDS)	Bulk formulation/capsule	Fenofibrate, *etc.*	Rapid and non-destructive drug quantification in complex lipid-based formulations and real-time monitoring of the production of SEDDS	Kuentz *et al.*[73]
Process monitoring (API dispersion)	Suspension	Not specified	Real-time evaluation of API dispersion homogeneity in the active premix process	Yonzon *et al.*[74]
Process monitoring (crystallization)	Solid/liquid	Mannitol	*In situ* qualitative monitoring of the polymorphic transformation of the metastable α-form to the stable β-form	Glennon *et al.*[75]
Process monitoring (wet granulation)	Tablet	Not specified	At-line measurement of polymorphic and/or pseudo-polymorphic phase transformation during the wet granulation step	Wang *et al.*[76]

Littlejohn's group studied the effect of particle size on the spectral features and Raman sampling depth for microcrystalline cellulose (Avicel), aspirin and sodium nitrate.[63–65] The particle size, shape, density and/or light absorption capability of the compounds were examined to find the minimum Raman sampling depth when monitoring the addition of aspirin to Avicel in the blender. The analytical performance of the proposed Raman method was compared with those of two other non-invasive monitoring techniques, near-infrared (NIR) spectrometry and broadband acoustic emission spectrometry. In addition, using *in situ* Raman measurement based on the WAI scheme, they investigated the evaporation of methanol from needle-shaped particles of cellobiose octaacetate (COA) in a jacketed vacuum drier and finally concluded that Raman spectroscopy is a valuable analytical method capable of monitoring the progress of drying even under agitation of particles.

Raman spectroscopic monitoring of the crystallization process was readily exploited by Feth's group.[66,67] Initially, they developed a new dynamic water vapor sorption gravimetry (DWVSG)–Raman spectroscopy coupled system and used it to investigate water (de)sorption-induced solid-phase transition of APIs (SAR474832). They showed that wide area coverage was appropriate for the analysis since the energy density per unit area was lower, so there was no influence on the gravimetrical data or on the samples such as phase transformations or even degradation by laser excitation. With the developed system, they confirmed that three different phases, the non-stoichiometric hydrate phase (form 1), crystallographic phase (form 2) and intermediate crystal phase (form 3), were involved in the rehydration process. Then, they moved the same analytical system to a scale-up pilot plant to monitor the crystallization process of SAR474832 in an on-line manner. On-line Raman measurement

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clearly probed a phase transition from forms 1 to 2 at ~65 °C followed by melting of the sample at ~105 °C.

Roggo/Dégardin's group successfully utilized the WAI scheme in the field of forensic science[68,69] for the identification of 31 types of genuine pharmaceutical tablets (25 samples) and capsules (6 samples), and detection of their counterfeits (27 samples). In their study, a support vector machine (SVM) was used as a classification method. When a counterfeit belonged to an existing class, no further analysis was performed to determine its chemical composition. When a new class of counterfeits occurred, the corresponding compositional analysis was performed using Raman spectroscopy, GC-MS and IR spectroscopy for further chemical profiling with the aid of principal component analysis (PCA) and correlation distance measurements.

Moffat's group compared the performance of laboratory and handheld Raman instrumentation for identification of counterfeit medicines, including 200 tablets containing a total of 42 different APIs obtained from 11 countries.[70] Obviously, the laboratory-based instrument showed better spectral quality; however, they demonstrated that the handheld instruments were potentially viable in future since a rapid, simple and cost-effective analytical method becomes more demanding.

In addition to the above-listed research groups, several researchers have also utilized Raman measurement based on wide area coverage for pharmaceutical process monitoring[71–76] such as the quantitative assay in self-emulsifying drug delivery systems (SEDDS),[73] real-time evaluation of API dispersion homogeneity in the active premix process,[74] *in situ* monitoring of polymorphic transformation[75] and at-line measurement of API transformation.[76] The plethora of uses for the WAI scheme directly indicates its superior analytical abilities necessary for reliable analysis of pharmaceutical samples including its improved spectral representation of samples and insensitivity to minor variations of the sample position with respect to the focal plane. As demonstrated, wide area coverage Raman spectroscopy will play a crucial role in the future as a viable tool for PAT in pharmaceutical industries.

(b) Applications in analysis of polymer/chemical/petrochemical samples. Table 2 summarizes applications of the WAI scheme for analysis of polymer/chemical/petrochemical samples as well as others. As shown, expansion of the applications was largely driven by Chung's group. They wanted to demonstrate the utility of WAI-based Raman analysis for samples that had also been analyzed with NIR spectroscopy in order to promote a healthy competition between NIR and Raman spectroscopy as fast, non-destructive analytical tools.

Initially, they compared Raman and NIR spectroscopy to determine the density of polyethylene (PE) pellets.[44] Raman and diffuse reflectance NIR spectra were collected for the packed PE pellets in a glass vial, and PLS was used to determine the density of the samples using both spectra. The prediction error was 0.0008 g cm^{-3} using Raman spectroscopy and 0.0011 g cm^{-3} using NIR spectroscopy. The harmonization of more selective Raman features and a reproducible spectral collection using the WAI scheme led to better accuracy of density determination.

Then, they studied a strategy to improve the accuracy of PE density determination by optimizing the sample temperature for Raman measurement.[45,46] For this purpose, Raman spectra of the same PE pellets were collected at eight different temperatures ranging from 30 to 100 °C in 10 °C intervals, and the accuracies achieved at the different temperatures were compared. The most accurate determination of PE density was achieved at 70 °C. The pre-melt behavior of PEs occurring around 70 °C enhanced Raman spectral selectivity among the samples. The enhanced spectral selectivity was responsible for the improved accuracy. In addition, using two-dimensional (2D) correlation analysis, they investigated the thermal behavior of PE below its melting temperature in relation to density variation.

Further, they also demonstrated a unique on-line Raman measurement of PE pellet density when the pellets were flying in a sample line.[47] While in flight, the pellets were sparsely populated, and Raman spectral collection covering a large spatial volume (larger number of pellets simultaneously) was necessary. Also, the measurement must be less sensitive to the pellet position, because the position and distribution were uncontrollable under flying conditions. For this purpose, the WAI scheme was used for spectral collection, and the resulting prediction error under flying conditions was comparable to that acquired when the pellets were packed. When a conventional Raman scheme covering a smaller sample volume with a short focal length was used, the peak intensity decreased dramatically, and the resulting signal-to-noise ratio was insufficient for reliable spectroscopic analysis.

Chung's group also compared the analytical performances of NIR and WAI-based Raman measurement for the compositional analysis of a metal aluminum etchant (MAE, a mixture of H_3PO_4, HNO_3, and CH_3COOH) solution,[53] one of the most frequently used etchant solutions. NIR and Raman spectra of circulating MAE solutions were simultaneously collected through the same Teflon (perfluoroalkoxy fluorocarbon, PFA) tube to ensure fair comparison. Further, they used Teflon tubing as an effective signal correction standard as well as a sample container (cell) without further modification of the WAI scheme.[51–54] The resulting Raman spectrum was the sum of spectral features from the sample and the Teflon tubing, which allowed the use of non-overlapping Teflon bands to correct for variations in the Raman intensity. The same WAI Raman measurement was also applied for the compositional analysis of a diluted sulfuric acid/hydrogen peroxide (DSP) solution,[52] a solvent for the removal of polymers in wet processes. They demonstrated that Raman spectroscopy is a suitable choice for on-line monitoring of etchant solutions, since many etchant solutions contain hydrogen peroxide and inorganic acids that have distinct Raman features. They also demonstrated the compositional analysis (total paraffin, total naphthene, and total aromatic content) of simulated naphtha samples using WAI-based Raman spectroscopy.[49] Overall, the prediction results using Raman spectroscopy were superior to those from NIR spectroscopy. In addition, they used the same system to determine ethanol and isopropanol concentrations.[51]

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Table 2 Summary of applications of the WAI scheme in Raman spectroscopic analysis for polymer/chemical/petrochemical and other samples

Area	Sample	Purpose of analysis	Analytical feature	Reference
Polymer	Polyethylene (PE) pellet	Determination of PE pellet density	Compared the accuracies of NIR and Raman spectroscopy for density determination	Chung *et al.*[44]
Polymer	PE pellet	Determination of PE pellet density	Improved Raman accuracy to determine the density by finding a measurement temperature providing improved spectral selectivity	Chung *et al.*[45]
Polymer	PE pellet	Study of the pre-melting behavior	Raman spectroscopy combined with two-dimensional (2D) correlation spectroscopy to examine the pre-melting behaviors of PE samples	Chung *et al.*[46]
Polymer	Flying PE pellet	Determination of PE pellet density	Direct on-line measurement of flying PE pellets in a sample line	Chung *et al.*[47]
Polymer	Organic/inorganic hybrid gel	Investigation of chemical bonding	Investigation of chemical bonding for polymerization under various synthesis conditions	Chung *et al.*[48]
Chemical	Etchant solution	Quantitative compositional analysis	Demonstrated a feasibility for measurement through Teflon tubing as an external standard as well as a sample cell	Chung *et al.*[52]
Chemical	Ethanol and isopropanol solutions	Determination of each concentration	Used Teflon as an intensity correction standard as well as the sample container for chemical analysis	Chung *et al.*[51]
Chemical	Etchant solution	Quantitative compositional analysis	On-line compositional monitoring directly through a Teflon tube and compared with the performance of NIR measurement	Chung *et al.*[53]
Petrochemical	Synthetic naphtha	Quantitative compositional analysis	More accurate prediction for the analysis of naphtha composition and compared with that resulted from NIR spectroscopy	Chung *et al.*[49]
Chemical	Residual volatiles in electrical cable	On-line monitoring of the quality of cable	Improved accuracy for monitoring the quality of final cable products	Kemper *et al.*[77]
Chemical	Sodium dodecyl sulfate	Investigation of the compression behavior	Investigation of the chemical structure change of the sample under pressure	Chen *et al.*[78]
Chemical	Polybromides	Process monitoring	*In situ* monitoring of bromination and debromination processes	Viana *et al.*[79]
Agriculture	Rice	Identification of rice samples with different geographical origins	Better compositional representation of rice grain samples and improved accuracy for determination of their origins	Chung *et al.*[50]
Gemology	Pearl	Discrimination of three different cultured pearls	Reproducible spectral collection of pearls with various shapes and sizes	Chung *et al.*[80]
Cosmetics	TiO_2-containing cream	Determination of TiO_2 concentration	Identified aggregation of TiO_2 particles in a cream and demonstrated the necessity of wide area coverage for reliable quantitative analysis	Chung *et al.*[81]

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Table 2 (*Contd.*)

Area	Sample	Purpose of analysis	Analytical feature	Reference
Clinical	Urea in dialysate solution	Determination of urea concentration	Increased the sensitivity using a copper reflector inside the Teflon tube	Chung *et al.*[54]
Clinical	Animal/human bone tissue	Assessment of bone quality	Investigation of the depth recovery of spectra of buried components and transcutaneous Raman spectral collection of bone	Morris *et al.*[82–91]
Environment	Paraquat	Improvement in reproducibility of SERS measurement	Prepared Au nanoparticle-encapsulated hydrogels as a SERS substrate and used the WAI scheme to improve the reproducibility	Chung *et al.*[92]

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In addition to the reports from Chung's group, several other researchers have shown applications of the WAI scheme in chemical analysis. The residual products of the cross-linking process for electrical cable materials, cumyl alcohol (CA) and acetophenone (AC) were measured using wide area coverage Raman spectroscopy.[77] Monitoring the level of byproducts was important since it directly influenced the dielectric properties of the insulation in the cable. The study showed that wide area coverage Raman spectroscopy was able to determine the concentrations of CA and AC reproducibly. Chen *et al.*[78] studied anionic polyhalides using Raman spectroscopy and found the existence of discrete monovalent pentabromides, heptabromides, and nonabromides. Viana *et al.*[79] elucidated the compression behavior of a powdered surfactant, sodium dodecyl sulphate (SDS), based on a comparison with the main component of a detergent formulation, *i.e.*, the chorine provider (sodium dichloroisocyanurate, DCCNa).

(c) Applications in other fields. WAI-based Raman measurement was used to improve the differentiation of rice samples from two different geographical origins.[50] For the purposes of comparison, a conventional scheme in which the laser illuminated only a small area was also used. The WAI scheme exhibited improved discrimination of the rice samples, primarily due to the acquisition of Raman spectra with more reliable sample representation as well as improved reproducibility, thereby demonstrating good potential for the analysis of diverse granular type agricultural samples, in which diffuse reflectance NIR spectroscopy has been popular.

With the use of discriminant-PLS, Raman spectroscopy was successfully used to discriminate three different groups of cultured pearls (fresh water, Akoya and South seawater).[80] The difference in the intensities of the $CaCO_3$ bands of Akoya and South seawater pearls provided a valuable clue for the discrimination. Although the pearls were hard-surfaced, round, solid samples of different sizes and surface shapes, a reproducible Raman spectral collection achieved using the WAI scheme played an important role in determining the pearl groups.

The concentration of TiO_2 in a cream (sun-block cream) formulation has been non-destructively measured using Raman spectroscopy without further sample pretreatments.[81] The distribution of TiO_2 particles in a highly viscous cream was not homogeneous on a microscopic scale and local aggregation of the particles could occur; therefore, acquisition of Raman spectra capable of representing the whole sample identity rather than the localized chemical information was critical to ensure reliable quantitative analysis. When Raman spectra were collected for the same sample using a conventional scheme with a much smaller laser spot (~100 μm), spectral reproducibility was degraded, as expected. When the WAI scheme was used, the resulting error in the determination of the TiO_2 concentration was 0.03 wt%, while it was 0.17 wt% when using the conventional scheme.

The urea concentration in an aqueous solution was determined by combining the WAI scheme and a Teflon tube as an effective intensity correction standard as well as a sample container.[54] To increase sensitivity, a copper reflector was positioned inside the Teflon tube to maximize the collection of Raman scattering. The obtained accuracy was 0.53 mM, close to the range of accuracy achieved in previous NIR studies (0.15–0.52 mM).

Morris' group extensively explored the potential of WAI-based Raman spectroscopy in the clinical and medical fields.[82–91] Initially, they developed a fiber-optic probe with uniform global illumination and a bundle of fifty 60 μm-core collection fibers arranged in a close-packed circle to collect Raman scattering from the illuminated region of samples. With the instrumentation, they profoundly investigated the depth recovery of spectra of buried components and subsurface mapping using several engineered polymers (polyethylene, Delrin, and Teflon).[82–84] Then, they expanded it for transcutaneous spectroscopy to examine canine[84] or chicken[85] tibiae below the skin surface, as a model system for human subject bone tissue spectroscopy and mapping. They concluded that Raman measurement of bone at the distal radius and other sites was feasible, and could be used for prediction of osteoporotic fragility fracture susceptibility.[87] Further, *in vivo* transcutaneous Raman spectral collections of bones[88] and the potential of Raman spectroscopy linked with computerized tomography (CT) imaging[89] in diverse medical examinations were successfully demonstrated by the group.

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Uniquely, the WAI scheme was also applied to improve the reproducibility of surface-enhanced Raman scattering (SERS) measurement.[92] SERS is a highly localized phenomenon, so the acquired Raman intensity may vary greatly depending on where the analyte is located on the SERS-generated nanostructure. Since the size, shape and distribution of nanostructures in a substrate are variable, individual SERS intensities arising from analytes differ substantially.[93] In this situation, when only a tiny area is sampled for SERS acquisition, the resulting intensity will fluctuate depending on where the laser illuminates the substrate. This is a crucial factor that affects the reproducibility of SERS-based quantitative analysis. The acquisition of SERS spectra by covering a large area of a substrate could effectively alleviate the problem of SERS intensity fluctuation. Gold nanoparticle-encapsulated hydrogels were used as a SERS substrate to detect low concentrations of paraquat (*N*,*N*′-dimethyl-4,4′-bipyridinium dichloride), a typical herbicide. When a microscope was used, the resulting RSD was 15.3%, while it was 2.5% with the use of the WAI scheme.

Conclusions

Raman measurement based on wide area coverage has been highly effective for enhancing sample representation and reproducibility in the spectral collection, and can eventually improve the accuracy of Raman spectroscopic analysis for diverse types of samples. Therefore, we believe that the applications of wide area coverage Raman spectroscopy will consistently increase in the future. Recently, transmission Raman spectroscopy and spatially offset Raman spectroscopy (SORS) have also emerged as effective methods to reduce the sub-sampling problem by also increasing the interaction of the laser radiation with the samples.[43,94–102] These methods are especially useful for the analysis of internally heterogeneous samples. Along with transmission Raman spectroscopy and SORS, wide area coverage Raman spectroscopy will be a major scheme in Raman analysis for QC/QA. A proper choice among the three methods needs to consider various factors, such as characteristics of samples, requirement of analysis time and signal-to-noise ratio of spectra, and will become more crucial for an optimal Raman spectroscopic analysis.

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